Redux & Stores: Professional State Management

MERN Stack Bootcamp Lecture
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Audience: Intermediate React Developers
Duration: ~90 minutes

🎯 Lecture Objectives

By the end of this session, you will be able to:

1. Understand **why Redux exists** and when it outperforms Context API & local state
2. Set up **Redux Toolkit (RTK)** using modern, industry-standard patterns
3. Create **slices**, **async thunks**, and **selectors** for scalable state management
4. Implement **RTK Query** for API caching, loading states, and automatic refetching
5. Architect a **production-ready Redux store** with proper folder structure & DevTools

> 💡 **Key Mindset:** Redux isn't a replacement for `useState`. It's a **predictable state container** for complex, cross-component data flows that require debugging, persistence, and strict update rules.

🤔 Why Redux? (The Professional Context)

State Tool	Best For	Limitations at Scale
`useState`	Local UI state (modals, inputs, toggles)	Doesn't share across components
Context API	Low-frequency global state (theme, auth, locale)	Triggers re-renders on every value change; no middleware
Redux/RTK	Complex app state, API caches, undo/redo, dev debugging	Boilerplate (mitigated by RTK), learning curve

When Teams Choose Redux

- ✅ Multiple components update the same data simultaneously
- ✅ You need **time-travel debugging** or state snapshots
- ✅ API responses must be cached, normalized, and shared across routes
- ✅ State updates require middleware (logging, analytics, error tracking, persistence)
- ✅ Large team needs predictable data flow & strict typing

🛠️ The Modern Stack: Redux Toolkit (RTK)

Legacy Redux (2015–2018) required manual switch statements, action creators, and immutable update helpers. **Today, 95% of professional codebases use Redux Toolkit**, the official, opinionated standard.

🔹 Step 1: Install & Configure Store

```
npm install @reduxjs/toolkit react-redux
```

```
// src/store/index.js
import { configureStore } from '@reduxjs/toolkit';
import cartReducer from './slices/cartSlice';
import authReducer from './slices/authSlice';

export const store = configureStore({
  reducer: {
    cart: cartReducer,
    auth: authReducer,
  },
  middleware: (getDefaultMiddleware) =>
    getDefaultMiddleware({ serializableCheck: false }), // Only disable if using non-serializable
data (rare)
  devTools: process.env.NODE_ENV !== 'production', // Auto-enabled in dev
});
```

🔹 Step 2: Wrap App with Provider

```
// src/main.jsx
import { Provider } from 'react-redux';
import { store } from './store';
import App from './App';

ReactDOM.createRoot(document.getElementById('root')).render(
  <Provider store={store}>
    <App />
  </Provider>
);
```

✅ **Professional Note:** `configureStore` automatically adds:

- Redux Thunk (async logic)
- Redux DevTools Extension
- Immutable state update checks
- Serializable state validation

📦 Core Building Blocks: Slices, Actions, Reducers

A **Slice** combines state, reducers, and actions in one file. This replaces the old 3-file pattern.

🔹 `src/store/slices/cartSlice.js`

```
import { createSlice } from '@reduxjs/toolkit';

const initialState = {
  items: [],
  total: 0,
  itemCount: 0,
};

export const cartSlice = createSlice({
  name: 'cart',
  initialState,
  reducers: {
    addItem: (state, action) => {
      const existing = state.items.find(i => i.id === action.payload.id);
      if (existing) {
        existing.quantity += 1;
      } else {
        state.items.push({ ...action.payload, quantity: 1 });
      }
    },
    removeItem: (state, action) => {
      state.items = state.items.filter(i => i.id !== action.payload);
    },
    clearCart: (state) => {
      state.items = [];
      state.total = 0;
      state.itemCount = 0;
    },
    // RTK uses Immer under the hood → "mutate" syntax is safe & clean
  },
});

export const { addItem, removeItem, clearCart } = cartSlice.actions;
export default cartSlice.reducer;
```

🔹 Consuming in Components

```
// src/components/CartPanel.jsx
import { useDispatch, useSelector } from 'react-redux';
import { addItem, removeItem, clearCart } from '../store/slices/cartSlice';

export default function CartPanel() {
  const dispatch = useDispatch();
  const { items, total, itemCount } = useSelector((state) => state.cart);

  return (
    <div className="bg-white p-4 rounded shadow">
      <h2>Cart ({itemCount} items)</h2>
      {items.map(item => (
        <div key={item.id} className="flex justify-between py-2">
          <span>{item.name} × {item.quantity}</span>
          <button onClick={() => dispatch(removeItem(item.id))}>✕</button>
        </div>
      ))}
```

```
      <p className="font-bold mt-3">Total: ${total.toFixed(2)}</p>
      <button onClick={() => dispatch(clearCart())}>Clear</button>
    </div>
  );
}
```

✅ **Why RTK Wins:** No `switch` statements, no manual object spreading, no action creator files. Just clean, predictable logic.

⚡ Async Data & RTK Query (Industry Standard)

Fetching data in Redux used to require `redux-thunk` + manual loading/error states. **RTK Query** now handles 80% of API logic automatically.

🔹 Option 1: Async Thunks (Custom Logic)

Use when you need fine-grained control over API calls, retries, or side effects.

```
// src/store/slices/userSlice.js
import { createSlice, createAsyncThunk } from '@reduxjs/toolkit';

export const fetchUser = createAsyncThunk('user/fetch', async (userId) => {
  const res = await fetch(`/api/users/${userId}`);
  if (!res.ok) throw new Error('Failed to fetch user');
  return res.json();
});

const userSlice = createSlice({
  name: 'user',
  initialState: { data: null, loading: false, error: null },
  reducers: {},
  extraReducers: (builder) => {
    builder
      .addCase(fetchUser.pending, (state) => { state.loading = true; })
      .addCase(fetchUser.fulfilled, (state, action) => {
        state.loading = false;
        state.data = action.payload;
      })
      .addCase(fetchUser.rejected, (state, action) => {
        state.loading = false;
        state.error = action.error.message;
      });
  },
});

export default userSlice.reducer;
```

🔹 Option 2: RTK Query (Auto-Caching & Deduping)

Use for 90% of API endpoints. Handles caching, loading, polling, and refetching automatically.

```
// src/store/api/productsApi.js
import { createApi, fetchBaseQuery } from '@reduxjs/toolkit/query/react';

export const productsApi = createApi({
  reducerPath: 'productsApi',
  baseQuery: fetchBaseQuery({ baseUrl: '/api' }),
  tagTypes: ['Product'],
  endpoints: (builder) => ({
    getProducts: builder.query({
      query: () => '/products',
      providesTags: ['Product'],
    }),
    addProduct: builder.mutation({
      query: (newProduct) => ({
        url: '/products',
        method: 'POST',
        body: newProduct,
      }),
```

```
      invalidatesTags: ['Product'],
    }),
  }),
});

export const { useGetProductsQuery, useAddProductMutation } = productsApi;
```

```
// Usage in component
function ProductList() {
  const {  products, isLoading, error } = useGetProductsQuery();
  if (isLoading) return <Spinner />;
  if (error) return <p>Failed to load</p>;
  return <div>{products.map(p => <ProductCard key={p.id}  {...p}  />)}</div>;
}
```

✅ **Professional Note:** RTK Query replaces `useFetch` custom hooks. It deduplicates requests, caches responses, and invalidates automatically.

🧬 Production-Grade Patterns

1. Normalized State Shape

Avoid nested arrays/objects. Store entities by ID for fast lookups & updates.

```
// ❌ Bad: Nested, hard to update
state: {
  categories: [
    { name: 'Electronics', products: [{ id: 1, name: 'Mouse' }, ...] }
  ]
}

// ✅ Good: Normalized (RTK `createEntityAdapter` helps)
state: {
  categories: { byId: { cat1: { id: 'cat1', name: 'Electronics' } }, allIds: ['cat1'] },
  products: { byId: { 1: { id: 1, name: 'Mouse', categoryId: 'cat1' } }, allIds: [1] }
}
```

2. Memoized Selectors (`createSelector`)

Prevent unnecessary re-renders when derived data hasn't changed.

```
import { createSelector } from '@reduxjs/toolkit';

const selectCartItems = (state) => state.cart.items;
const selectFilter = (state) => state.ui.filter;

export const selectFilteredCartItems = createSelector(
  [selectCartItems, selectFilter],
  (items, filter) => items.filter(i => i.category === filter)
);
// Only recomputes when items OR filter change
```

3. DevTools Integration

- Install Redux DevTools browser extension
- Track every action, state snapshot, and async thunk lifecycle
- Time-travel debug: revert to previous states instantly
- **Production tip:** Disable in prod via `devTools: process.env.NODE_ENV !== 'production'`

4. Error Boundary + Store Recovery

Wrap Redux state in error boundaries. On critical failure, reset to safe defaults:

```
import { useDispatch } from 'react-redux';
import { clearCart } from './store/slices/cartSlice';

function ErrorFallback({ resetErrorBoundary }) {
  const dispatch = useDispatch();
  const recover = () => {
    dispatch(clearCart());
    resetErrorBoundary();
  };
  return <button onClick={recover}>Reset Cart & Retry</button>;
}
```

📂 Scalable Folder Structure (Enterprise)

```
src/
├── store/
│   ├── index.js              # configureStore
│   ├── middleware/           # Custom middleware (analytics, logging, persist)
│   ├── slices/               # Feature-specific slices
│   │   ├── cartSlice.js
│   │   ├── authSlice.js
│   │   └── uiSlice.js
│   └── api/                  # RTK Query endpoints
│       ├── productsApi.js
│       └── ordersApi.js
├── features/                 # Co-located UI + state (optional but recommended)
│   ├── cart/
│   │   ├── CartPanel.jsx
│   │   └── cartSelectors.js
│   └── auth/
│       ├── LoginForm.jsx
│       └── authUtils.js
└── hooks/
    └── useStore.js           # Typed useSelector/useDispatch wrappers (TS)
```

✅ **Why this works:** Scales to 50+ slices. Co-location keeps related code together. API layer stays separate from UI state.

🚫 When NOT to Use Redux

Scenario	Better Alternative
Form state	`react-hook-form` + local state
Theme/Auth (low frequency)	Context API
Simple toggle/modals	`useState`
Real-time chat/cursors	`zustand`, `jotai`, or WebSockets
Server data only	`@tanstack/react-query` + RTK Query

> 💡 **Rule of Thumb:** If your state doesn't cross 3+ components, require complex update logic, or need DevTools debugging, don't use Redux. Start simple, scale intentionally.

🎯 Instructor Closing

Redux isn't magic. It's **predictability**. When you standardize how state changes, you unlock:

- Instant debugging via DevTools
- Time-travel & state snapshots
- Clean separation of UI, logic, and data fetching
- Team-wide patterns that survive onboarding

> 💬 **Your Next Step:**
> Take one feature from your current project that uses prop drilling or messy `useEffect` chains. Extract it into a slice or RTK Query endpoint. Add DevTools. Test it. You'll never want to go back to unmanaged global state.
>
> Tomorrow's lab will have you refactor the POS cart from prop-drilling to Redux. We'll cover persistence, selectors, and error boundaries.
>
> Push your repos, tag `#MERNBootcamp-Redux`, and I'll review your slice architecture & async patterns. Keep state predictable, actions explicit, and selectors memoized.
>
> — Kashif Raza | Structured state, scalable apps 🛒
